

UNIVERSITY OF BRISTOL

School of Mathematics

FINANCIAL MATHEMATICS

MATH35400

(Paper code MATH-35400)

MAY / JUNE 2022 2 hours 30 minutes

This paper contains 4 questions. All answers will be used for assessment.

Calculators of an approved type (permissible for A-Level examinations) are permitted.

Candidates may bring four sheets of A4 notes written double-sided into the examination.

On this examination, the marking scheme is indicative and is intended only as a guide to the relative weighting of the questions.

Do not turn over until instructed.

1. **(25 marks)** Consider a single-period model with N risky stocks and $\Omega = \{\omega_1, \omega_2, \dots, \omega_K\}$. Let $S_n(t)(\omega)$ denote the value of the n th stock at time point t in state ω . Assume there is a bank account with interest rate r .

- (a) **(3 marks)** Explain the interpretation of a trading strategy $H = (H_0, H_1, \dots, H_N)$. In particular, what does it mean if H_i is negative for some $i \geq 0$?
- (b) Assume that $N = 2$, $K = 3$ and $r = 0.2$ and that the price processes of the two stocks are given by:

n	$S_n(0)$	$S_n(1)(\omega_1)$	$S_n(1)(\omega_2)$	$S_n(1)(\omega_3)$
1	10	14.4	10.8	12
2	C	15.6	14.4	9.6

- i. **(6 marks)** For an arbitrary trading strategy H , determine the value process V and gains process G , and their discounted versions V^* and G^* .
- ii. **(7 marks)** For $\mathbb{Q} = (q_1, q_2, q_3)$, state a system of equations that the q_i must satisfy in order for $\mathbb{Q}$ to be a risk-neutral probability measure for this model. Determine for which values of C there exists a solution to this system of equations. For values of C too small for a solution to exist, explicitly describe an arbitrage strategy.
- iii. **(3 marks)** If $C = 10$, use the risk-neutral valuation principle to find the fair prices of:
- a European call option with strike price £13 based on the first stock,
 - a European put option with strike price £13 based on the second stock.
- (c) **(6 marks)** Let $0 < d < 1 < u$ and consider a model with just one stock, ie $N = 1$. Assume that the value of the stock at time point 0 is S_0 . At time point 1, the value of the stock is either $S_1 = dS_0$ or $S_1 = S_0$ or $S_1 = uS_0$, each with strictly positive probability p_1, p_2 and p_3 respectively.
- Find values of r for which the model is free of arbitrage.
 - Find values of r for which the model is complete.

You need to prove/justify your answers.

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2. (25 marks)

- (a) The current price for Bibi shares is £50 per share. A put option on Bibi shares maturing in exactly 9 months with a strike price of £40 is currently trading at £14. The price of the corresponding call option with the same strike price and same maturity is currently £26.
- i. (4 marks) The put-call parity states that if there are no arbitrage opportunities, the following formula holds:

$$S_t + P_t - C_t = Ke^{-r(T-t)}.$$

State the definition of each of the quantities appearing in this formula and use it to show that when $r = 0.05$ and time t is measured in years, there exist arbitrage opportunities in the model above.

- ii. (7 marks) Describe explicitly a trading strategy which will give riskless profit.
- (b) Consider the following model with two periods assuming that the interest rate is $r = 0$ and assume that u and d are arbitrary constants with $u > 6 > d$. We assume that at each point of time t an investor is aware of all prices up to that time point.

State	$t = 0$	$t = 1$	$t = 2$
ω_1	10	14	16
ω_2	10	14	12
ω_3	10	6	u
ω_4	10	6	d

An American *look-back put option* is an option where the holder is paid the amount $Y_t(\omega)$ where

$$Y_t(\omega) = (\max_{\nu: \nu \leq t} S_\nu(\omega) - S_t(\omega))^+.$$

- i. (4 marks) Calculate the risk-neutral probability measure $\mathbb{Q}$ and specify $\mathbb{Q}(\{\omega_i\})$ for every $i = 1, 2, 3, 4$.
- ii. (2 marks) Assume that $u = 11$ and $d = 5$ and determine the payoff Y_t of the American look-back put option for every state and every time point.
- iii. (5 marks) Use risk-neutral valuation to find the time $t = 0$ value of the option assuming that $u = 11$ and $d = 5$.
- iv. (3 marks) Decide whether an early exercise of the option at time point 1 is optimal in the case where $10 > u > 6 > d$.

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3. (25 marks)

- (a) Consider a Cox–Ross–Rubinstein process with 2 time periods, where the upward and downward steps are given by $u = 1.2$, $d = 0.9$ respectively and $S_0 = 60$. The interest rate is $r = 0.11$, and the bank account process grows like $(1 + r)^t$.
- (2 marks) Construct a tree of the possible values taken by this process.
 - (2 marks) Check that a unique equivalent martingale measure $\mathbb{Q}$ exists and state the form of $\mathbb{Q}$.
 - (5 marks) Hence, using the risk-neutral valuation principle, price an option with a payoff $\left(\sum_{i=1}^T S_i - K\right)^+$ for $K = 120$ and expiry time $T = 2$.
- (b) You may assume that for an Itô process of the form $dX_t = b_t dt + \sigma_t dW_t$, any sufficiently well-behaved function $f(t, X_t)$ satisfies Itô's formula

$$df = \left[f_t + b_t f_x + \frac{1}{2} \sigma_t^2 f_{xx} \right] dt + \sigma_t f_x dW_t.$$

- (5 marks) List the conditions that a process W_t must satisfy to be a standard Brownian motion. Prove that cW_{t/c^2} is also a standard Brownian motion by showing that it satisfies the axioms.
- (5 marks) Using Itô's formula, state a condition involving the derivatives of f under which $f(t, X_t)$ is a martingale.
Deduce conditions on constants A , B and function $g(t)$ under which the process $Y_t = AW_t^2 + BW_t + g(t)$ is a martingale.
- (6 marks) Consider the stopping time T when Brownian motion W_t first hits the set $\{1, -1\}$ (so W_t^2 is first equal to 1).
By considering two processes of the form $Y_t = AW_t^2 + BW_t + g(t)$ find $\mathbb{P}(W_T = 1)$ (and explain why this value is not a surprise) and $\mathbb{E}T$.
Hint: You may assume that for any martingale of the form Y_t and stopping time τ , the $\mathbb{E}Y_\tau = \mathbb{E}Y_0$.

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4. **(25 marks)** Let $(W_t)_{t \geq 0}$ be a standard Brownian motion. The Black-Scholes model is a financial market model in continuous time with two assets: A riskless bond with constant interest rate r and a stock whose price S_t satisfies the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

for constants $\mu \in \mathbb{R}$, $\sigma > 0$.

- (a)
 - i. **(2 marks)** What name is given to the parameter σ ? Is this parameter assumed to be known or unknown?
 - ii. **(1 mark)** Write S_t in terms of W_t
 - iii. **(3 marks)** Let $\mathbb{Q}$ be the equivalent martingale measure. Define the process $\bar{W}_t$ such that it is a standard Brownian motion under $\mathbb{Q}$. Write S_t in terms of $\bar{W}_t$.
- (b) For a given positive integer α , consider a contingent claim which pays $X(\alpha) = (S_T)^\alpha$ at time T .
 - i. **(6 marks)** Using the risk-neutral valuation principle calculate the arbitrage-free price $P(t, S_t)$ of this claim at time t in terms of the stock price, S_t , at time $0 \leq t \leq T$. [Hint: For this question you may use that the moment-generating function of a Gaussian random variable X with mean μ and variance σ^2 is $M_X(t) = \exp(\mu t + \sigma^2 t^2/2)$.]
 - ii. **(5 marks)** The Black-Scholes-Merton differential equation is

$$f_t + r x f_x + \frac{1}{2} \sigma^2 x^2 f_{xx} - r f = 0.$$

What is the role of f and what boundary conditions does it need to satisfy?

Assume that $\alpha = 2$ and show that the corresponding price function $P(t, S_t)$ from (b i) satisfies the BSM equation.

- iii. **(2 marks)** Suppose that V_0 is the time 0 price that the option is trading at. Explain what is meant by the ‘implied volatility’ of S_t .
- iv. **(6 marks)** Consider the Black-Scholes model with interest rate $r = 0.1$. Suppose that the contingent claim with payoff $X(2) = (S_1)^2$ (i.e. taking $\alpha = 2$ and $T = 1$ above) is trading for £120 at time 0, when the stock has value $S_0 = 10$. Deduce the value of the implied volatility of S_t .
Assuming this implied volatility is the true value of σ , deduce the time zero value of the contingent claim with payoff $X(3) = (S_1)^3$ based on the same stock.

End of examination.